

Prediabetes: A high-risk state for developing diabetes

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Summary Prediabetes (or “intermediate hyperglycaemia”), based on glycaemic parameters above normal but below diabetes thresholds is a high risk state for diabetes with an annualized conversion rate of 5%–10%; with similar proportion converting back to normoglycaemia. The prevalence of prediabetes is increasing worldwide and it is projected that >470 million people will have prediabetes in 2030. Prediabetes is associated with the simultaneous presence of insulin resistance and β -cell dysfunction, abnormalities that start before glucose changes are detectable. Observational evidence shows associations of prediabetes with early forms of nephropathy, chronic kidney disease, small fibre neuropathy, diabetic retinopathy, and increased risk of macrovascular disease. Multifactorial risk scores could optimize the estimation of diabetes risk using non-invasive parameters and blood-based metabolic traits in addition to glycaemic values. For prediabetic individuals, lifestyle modification is the cornerstone of diabetes prevention with evidence of a 40%–70% relative risk reduction. Accumulating data also suggests potential benefits from pharmacotherapy.